SCHOOL DATABASES

1. Create Operation:

```
C:\Users\parja>mongosh
Current Mongosh Log ID: 69f5c93e6d0662d8eb3682d0
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serve
Using MongoDB:          8.2.7
Using Mongosh:          2.8.2
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-04-30T18:32:11.440+05:30: Access control is not enabled for the databa
 is unrestricted
------

test> show databases  //show how many db present
admin      40.00 KiB
config    104.00 KiB
libraray    8.00 KiB
library    60.00 KiB
local      40.00 KiB
school     68.00 KiB
test> use school //switched to school db
switched to db school
school> show collections //show how many collections in school db
students
school> db.school.drop() //delete collection
true
school> db.createCollection("student"); //create collection
{ ok: 1 }
```

2. Insert Data:

```
school> db.students.insertMany([
|   { name: "Millie", age: 22, course: "Data Analyst", status: "ongoing" },
|   { name: "Gwen", age: 21, course: "MERN Stack", status: "ongoing" },
|   { name: "Parker", age: 23, course: "Java", status: "completed" },
|   { name: "Conrad", age: 22, course: "MERN Stack", status: "ongoing" },
|   { name: "Belly", age: 24, course: "Data Science", status: "completed" }
| ]); // insert data
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f5cc3c6d0662d8eb3682db'),
    '1': ObjectId('69f5cc3c6d0662d8eb3682dc'),
    '2': ObjectId('69f5cc3c6d0662d8eb3682dd'),
    '3': ObjectId('69f5cc3c6d0662d8eb3682de'),
    '4': ObjectId('69f5cc3c6d0662d8eb3682df')
  }
```

3. Read data:

```
school> db.students.find() // read data
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682db'),
    name: 'Millie',
    age: 22,
    course: 'Data Analyst',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cd6e6d0662d8eb3682e0'),
    name: 'Hary',
    age: 23,
    course: 'python',
    status: 'Completed'
  }
]
```

4. Delete data:

```
school> db.students.deleteMany(
| {name: "Hary"}) //delete data
{ acknowledged: true, deletedCount: 1 }
school> db.students.find()
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682db'),
    name: 'Millie',
    age: 22,
    course: 'Data Analyst',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  }
]
```

5. **Update data:**

```
school> db.students.updateMany(
| {name: "Millie"}, {$set:{status: "completed"}}
| ) //update the data
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
school> db.students.find()
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682db'),
    name: 'Millie',
    age: 22,
    course: 'Data Analyst',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  }
]
```

6. **Greater than and Lesser than Operation:**

```
school> db.students.find({ age: { $gt: 22 } })
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  }
]
school> db.students.find({ age: { $lt: 22 } })
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
```

7. **In Operation:**

```
school> db.students.find({ course: { $in: ["MERN Stack", "Java"] } })
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
```

8. **And Operation:**

```
school> db.students.find({
|   $and: [
|     { course: "MERN Stack" },
|     { status: "ongoing" }
|   ]
| })
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
```

9. Or Operation:

```
school> db.students.find({
|   $or: [
|     { age: 21 },
|     { status: "completed" }
|   ]
| })
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682db'),
    name: 'Millie',
    age: 22,
    course: 'Data Analyst',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  }
]
```

10. Exist Operation:

```
school> db.students.find({ age: { $exists: true } })
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682db'),
    name: 'Millie',
    age: 22,
    course: 'Data Analyst',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  }
```

11. Limit and Skip Operation:

```
school> db.students.find().limit(2)
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682db'),
    name: 'Millie',
    age: 22,
    course: 'Data Analyst',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
school> db.students.find().skip(2)
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  }
]
```

12. Ascending Operation:

```
school> db.students.find().sort({ age: 1 }) //ascending
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682db'),
    name: 'Millie',
    age: 22,
    course: 'Data Analyst',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  }
]
```

13. Descending Operation:

```
school> db.students.find().sort({ age: -1 }) //descending
[
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682df'),
    name: 'Belly',
    age: 24,
    course: 'Data Science',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dd'),
    name: 'Parker',
    age: 23,
    course: 'Java',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682db'),
    name: 'Millie',
    age: 22,
    course: 'Data Analyst',
    status: 'completed'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682de'),
    name: 'Conrad',
    age: 22,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('69f5cc3c6d0662d8eb3682dc'),
    name: 'Gwen',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
```